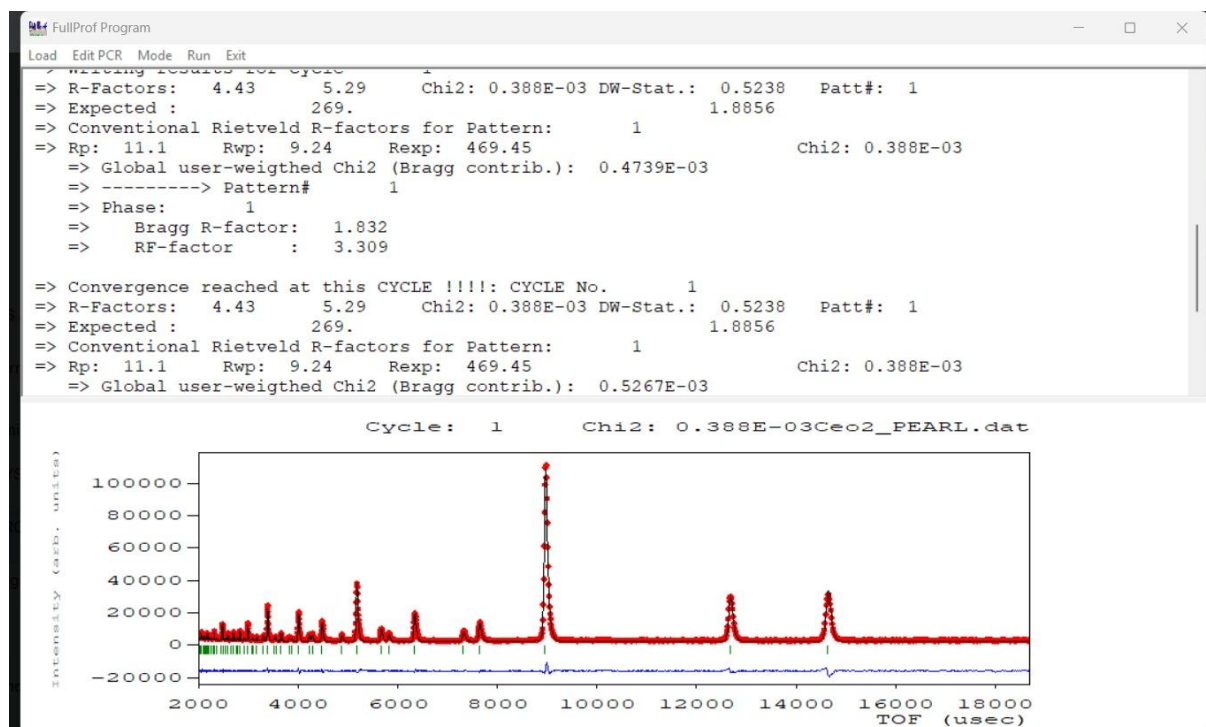


Exp. 12. Rietveld refinement & lattice parameter determination (single-phase material) using FullProf software

AIM:

To perform Rietveld refinement of X-ray diffraction (XRD) data and determine the lattice parameters of crystalline materials using FullProf software.

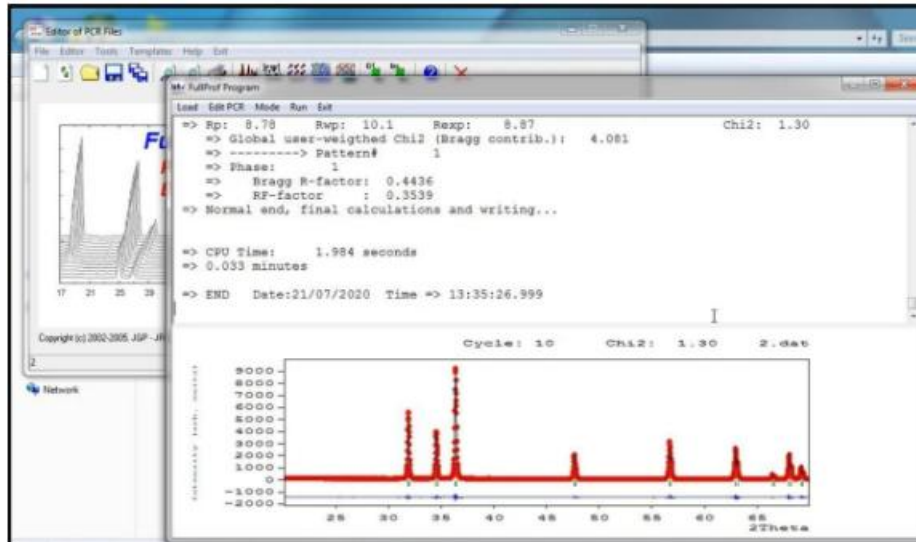
Case Study 1: Perform Rietveld refinement for a metal oxide material (CeO₂) using the pseudo-Voigt Peak Profile. Analyse the RWP Values



- This is the final Rietveld refinement output of the CeO₂ XRD data in FullProf.
- The graph shows the experimental and calculated XRD profiles, the blue difference line, and green Bragg peak positions.
- $R_p = 11.1\%$ represents the profile difference, while $R_{wp} = 9.24\%$ represents the weighted profile difference; lower values generally indicate better agreement.
- The refinement stopped at Cycle 1 because the convergence condition was reached, and “Normal end” confirms that the refinement completed successfully.

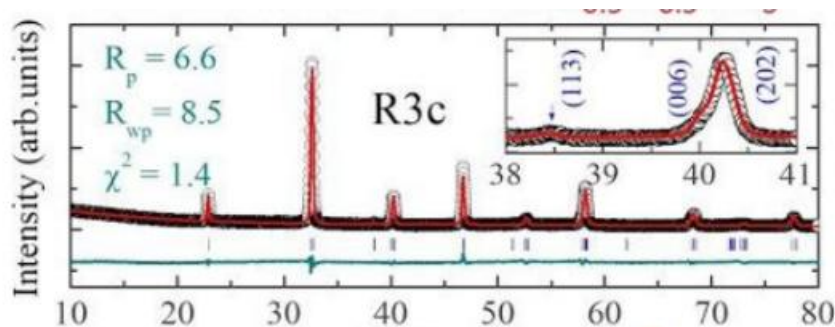
- The important final values are $R_p = 11.1\%$, $R_{wp} = 9.24\%$, $\chi^2 = 0.388 \times 10^{-3}$, Bragg R-factor = 1.831; therefore, the CeO_2 Rietveld refinement was successfully performed.

Case Study 2: Determination of lattice parameters for metal oxide material (ZnO).



- X-axis: The graph shows 2θ (two-theta angle), which is typical of an X-ray diffraction (XRD) pattern.
- Y-axis: The intensity of the diffracted X-rays is plotted on the vertical axis.
- Sharp peaks: Several sharp peaks are observed, indicating that the material has a crystalline structure.
- Peak positions: The different peaks at various 2θ values represent different crystal planes (hkl) in the material.
- Phase identification: The pattern can be compared with standard XRD data to identify the phases and determine the crystal structure of the sample.

Case Study 3: Determination of lattice parameters for doped metal oxide material ($\text{Na}_{0.5}\text{Bi}_{0.5}\text{TiO}_3$).



1. X-axis: The graph represents the 2θ diffraction angle, ranging approximately from 10° to 80° .
2. Y-axis: The vertical axis shows the XRD intensity in arbitrary units.
3. Sharp peaks: The strong and sharp peaks indicate that the sample has a well-crystalline structure.
4. Crystal phase: The pattern is identified with the R3c crystal structure, and the peaks are indexed to different crystal planes such as (113), (006), and (202).
5. Fitting quality: The experimental data and fitted curve show good agreement, indicated by $R_p = 6.6$, $R_{wp} = 8.5$, and $\chi^2 = 1.4$, suggesting a good-quality structural refinement.